

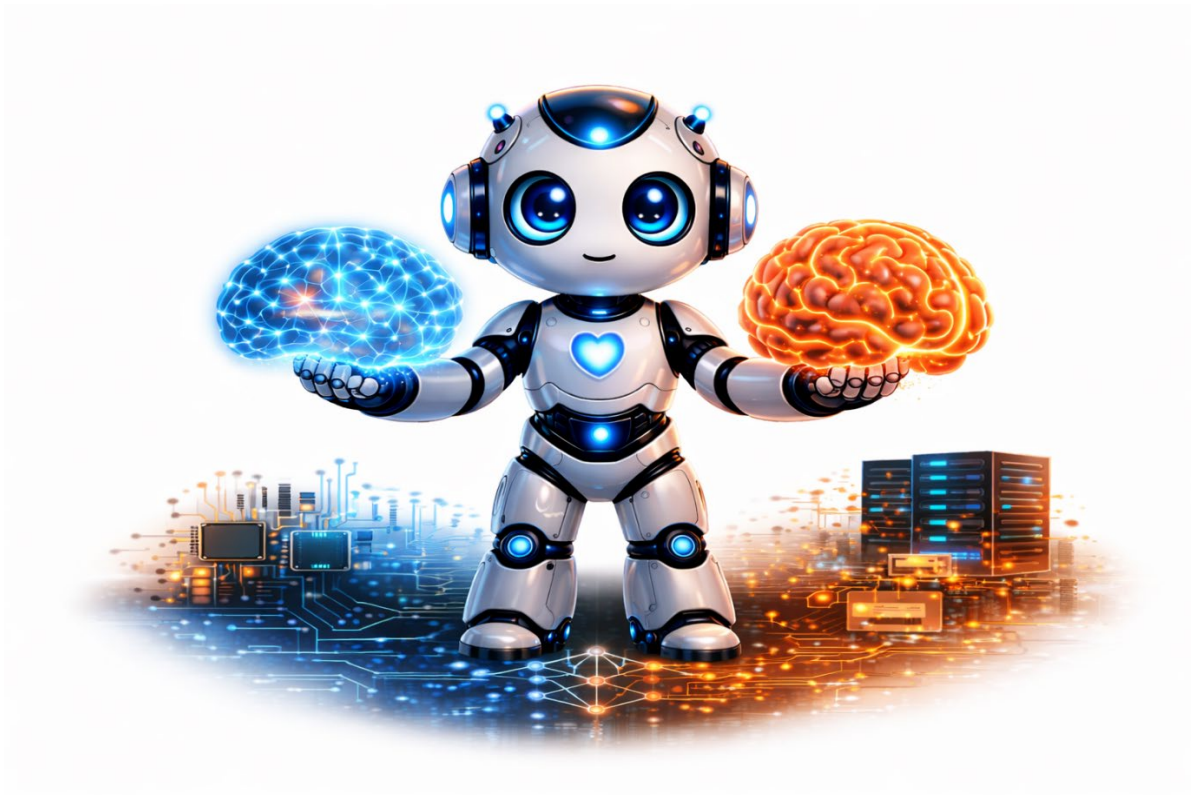
ITAI 4374: Neuroscience as a Model for AI

Module 06

Memory, Learning, and AI Architectures

From Remembering to Reasoning

Houston City College • AI & Robotics Program
Professor Patricia McManus



Section 1

Welcome — Why Memory Changes Everything

WHAT YOU WILL LEARN IN THIS MODULE

By the end of Module 06, you will be able to:

1. Explain the major biological memory systems (working, episodic, semantic, procedural) and identify which brain structures support each.
2. Describe how the hippocampus consolidates memories during sleep — and why this matters for AI.
3. Connect hippocampal replay to experience replay in Deep Reinforcement Learning.
4. Compare AI memory architectures (LSTMs, transformers, memory-augmented networks) to their biological inspirations.
5. Articulate the biological plausibility debate: why backpropagation is powerful but biologically unlikely, and what alternatives exist.
6. Evaluate which biological memory principles AI has successfully adopted and which remain open challenges.

Imagine waking up every morning with no memory of yesterday. You recognize your family, you can tie your shoes and speak your language, but you have no idea what happened last Tuesday. You cannot remember a single conversation, a single meal, a single moment from the past thirty years. This was the life of Henry Molaison — the most famous patient in the history of neuroscience, known for decades only as H.M.

In 1953, surgeons removed most of Henry's hippocampus to treat severe epilepsy. The seizures improved. But Henry lost something far more precious: the ability to form new memories. For the next fifty-five years, until his death in 2008, Henry lived in a permanent present. He met his doctors anew every single day. He read the same magazines without recognizing them. He could hold a conversation for a few minutes, but the moment he was distracted, the conversation vanished as if it had never happened.

We introduced Henry briefly in Module 03 when we discussed brain anatomy. Now we are going to understand why his case matters so deeply — not just for neuroscience, but for artificial intelligence. Henry's tragedy revealed that memory is not one thing. It is a

collection of systems, each with different brain structures, different rules, and different computational purposes. And these distinctions have directly shaped how we build AI systems that learn and remember.



WHERE WE ARE IN THE COURSE

Module 04 taught you how neurons fire and connect into spiking networks.

Module 05 showed you how those networks process sensory information from the world — vision, hearing, touch, taste, and smell.

Module 06 asks the next question: once the brain perceives something, how does it STORE that information, RETRIEVE it later, and LEARN from it?

The arc: CONNECT → SENSE → REMEMBER → ATTEND & AWARE (Module 07) → DECIDE (Module 08)



WHY SHOULD YOU CARE?

Memory is the biggest unsolved problem in AI. Current large language models like GPT-4 and Claude have a 'context window' — a fixed buffer that is erased between conversations. They do not truly remember you. They cannot learn from experience the way you do.

Companies like Google DeepMind, Meta AI, and Anthropic are actively researching how to give AI persistent, flexible memory. Understanding how the brain solves this problem gives you a blueprint for the next generation of AI architectures — and the job skills employers are looking for.

PART 1

How the Brain Remembers

Conceptual — No Math Required



Section 2

The Four Memory Systems — Not All Memories Are Created Equal

One of the most important discoveries in neuroscience is that memory is not a single ability. You use fundamentally different brain systems to remember what you had for breakfast (episodic memory), to know that Paris is the capital of France (semantic memory), to ride a bicycle (procedural memory), and to hold a phone number in your head while dialing (working memory). These systems can be damaged independently — a person can lose one while keeping the others perfectly intact.

This discovery has profound implications for AI. Most AI systems have only one kind of memory — a set of weight parameters learned during training, plus a temporary input buffer. The brain, by contrast, has specialized memory systems optimized for different tasks. Understanding these systems helps us see both what AI does well and what it still lacks.

2.1 Working Memory — The Brain's Scratchpad

Working memory is your ability to hold and manipulate a small amount of information right now, in the moment. When someone tells you a phone number and you repeat it to yourself while walking to find a pen, that is working memory. When you do mental arithmetic — say, 47 plus 38 — you hold the intermediate results (7 plus 8 equals 15, carry the 1) in working memory.

The key features of working memory are its limited capacity and its temporary nature. The classic estimate, from George Miller's famous 1956 paper, is that working memory holds about seven items (plus or minus two). More recent research by Nelson Cowan suggests the true capacity is closer to four items. Either way, working memory is tiny compared to long-term memory.

The primary brain region supporting working memory is the prefrontal cortex, the same area we discussed in Module 03 when we talked about the frontal lobe's role in planning

and decision-making. Neurons in the prefrontal cortex show sustained firing — they remain active during a delay period, literally holding information online through persistent neural activity. When you keep a phone number in mind, specific prefrontal neurons are firing continuously to maintain that representation.

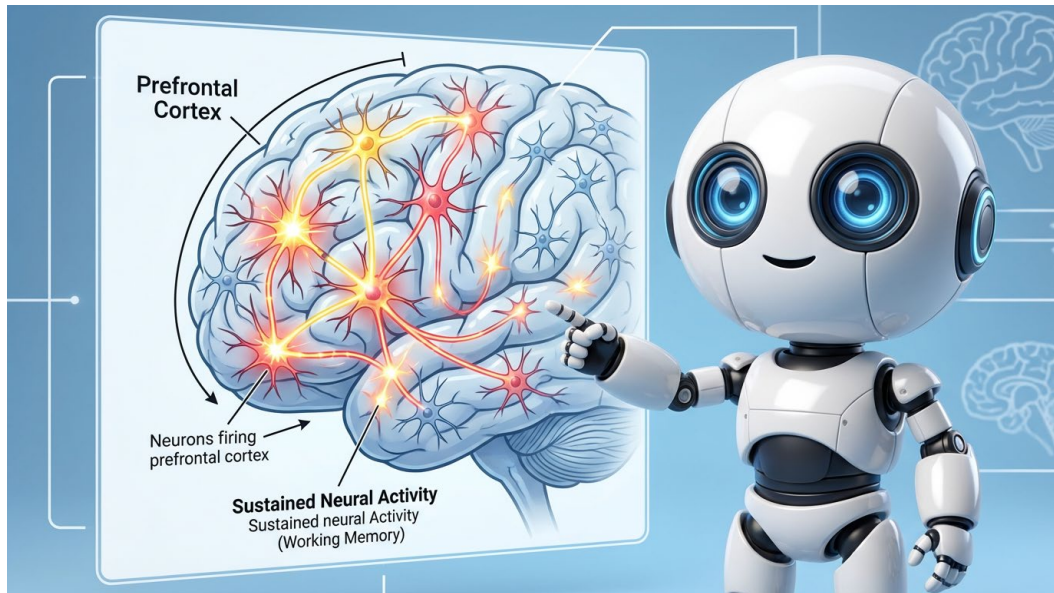


Figure 1: Prefrontal cortex sustained firing during a working memory delay task. Neurons fire continuously to maintain information.



AI CONNECTION: Working Memory in AI Systems

Working memory maps directly to several AI concepts:

Context window: In large language models (like GPT-4 or Claude), the context window is the text the model can 'see' at once. Like working memory, it is limited in size and temporary — once the conversation ends, it is gone.

Hidden state in RNNs: The hidden state vector that an RNN carries from one time step to the next is a form of working memory. It holds a compressed summary of everything processed so far. Like biological working memory, it has limited capacity and degrades over time (the vanishing gradient problem).

Registers in a CPU: At the hardware level, CPU registers hold the data currently being operated on — another analogy to working memory's role as the brain's active workspace.

Key insight: AI working memory is typically fixed in size, while biological working memory is flexible — you can focus more working memory on a difficult task. Attention mechanisms in transformers partially address this by dynamically allocating processing resources.

2.2 Episodic Memory — Your Personal Time Machine

Episodic memory is your ability to mentally travel back in time and re-experience events from your own life. It is the memory of 'what happened, where, and when.' When you remember your first day of college — the nervousness, the unfamiliar hallways, the person who sat next to you in your first class — that is episodic memory. It includes not just the facts of the event but the sensory details, emotions, and context.

The term 'episodic memory' was coined by Endel Tulving in 1972, and it is considered uniquely rich among memory types. It is also uniquely fragile. Episodic memory is the type most devastated by hippocampal damage — which is exactly what happened to Henry Molaison. After his surgery, Henry could not form new episodic memories. He could not remember meeting someone five minutes ago. But crucially, he could still learn new motor skills (procedural memory) and retain general knowledge from before his surgery (semantic memory). This dissociation proved that these memory systems are genuinely separate.

The hippocampus is the critical structure for forming episodic memories. It does not store the memories permanently — instead, it acts like an index or a binding agent, linking together the distributed cortical representations (the visual scene, the sounds, the emotions, the spatial location) that make up a complete episode. Over time, through a process called consolidation, these links are gradually strengthened in the cortex itself, and the hippocampus can eventually let go. This is why very old memories can survive hippocampal damage while recent ones cannot.

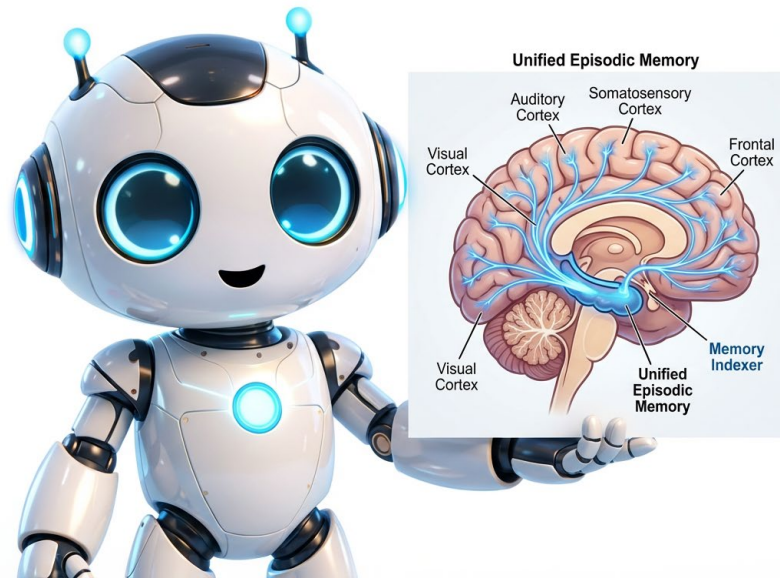


Figure 2: The hippocampus as a memory indexer — binding distributed cortical representations into a unified episodic memory.



AI CONNECTION: Episodic Memory in AI

Most AI systems have no true episodic memory. A language model does not 'remember' your previous conversation — it processes each session from scratch (unless the conversation is in context).

Some attempts to add episodic memory to AI:

- Retrieval-Augmented Generation (RAG): Stores past interactions in a database and retrieves relevant ones when needed. This is conceptually similar to the hippocampus indexing memories and retrieving them based on cues.
- Memory-augmented networks (like the Neural Turing Machine): Include an external memory bank the network can read from and write to.
- Episodic control in RL (MFEC, NEC): Store specific past experiences and use them directly for decision-making, rather than averaging everything into weights.

The gap: Biological episodic memory is automatic, context-rich, and emotionally tagged. AI 'episodic' systems require explicit engineering and lack the richness of human episodes.

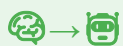
2.3 Semantic Memory — Knowledge Without Episodes

Semantic memory is your store of general knowledge about the world — facts, concepts, and meanings that are not tied to any specific personal experience. You know that Tokyo

is the capital of Japan, that water freezes at zero degrees Celsius, and that dogs are mammals. You probably do not remember the specific moment you learned any of these facts. The knowledge has been extracted from episodes and stored as abstract, context-free information.

The distinction between episodic and semantic memory was demonstrated in patients like Henry Molaison and others with hippocampal damage. While Henry could not form new episodic memories, he could still access semantic knowledge learned before his surgery. Interestingly, some patients with hippocampal damage can slowly acquire new semantic knowledge — they just cannot remember the learning episodes themselves. This suggests that while the hippocampus is essential for episodic memory, semantic memory can be supported by the cortex more directly.

Think of the relationship this way: episodic memory is like a diary — it records specific events with dates and contexts. Semantic memory is like an encyclopedia — it stores distilled knowledge without the personal narrative. Over time, many of your episodic memories get 'semanticized' — you lose the experiential details but retain the extracted facts.



AI CONNECTION: Semantic Memory = Model Weights

This is where AI actually excels. The weights of a trained neural network are essentially a form of semantic memory. During training, the network processes millions of examples (analogous to episodes) and distills them into abstract patterns stored in connection weights.

A language model 'knows' that Paris is in France — not because it remembers a specific training example, but because that knowledge has been extracted from thousands of text passages and encoded into weight patterns. This is strikingly similar to how semantic memory abstracts knowledge from episodes.

The parallel is almost exact: Training data = episodes. Learned weights = semantic memory. The process of training = consolidation from episodic to semantic.

The limitation: AI semantic memory is frozen after training (unless fine-tuned). Biological semantic memory updates continuously as you have new experiences.

2.4 Procedural Memory — The Body Remembers

Procedural memory is your memory for skills and habits — how to ride a bicycle, type on a keyboard, play a guitar chord, or tie your shoelaces. Unlike episodic and semantic memory, procedural memory is largely unconscious. You cannot easily describe in words how you balance on a bicycle; you just do it. If you try to consciously think about the mechanics while riding, you might actually perform worse. This phenomenon is sometimes called the 'centipede effect' — the centipede walked perfectly until someone asked which leg goes first.

Procedural memory is supported primarily by the basal ganglia (which we introduced in Module 03 as the brain's action selection system) and the cerebellum (the brain's motor coordinator). It is remarkably resilient — patients with severe amnesia who cannot remember yesterday can still learn new motor skills. Henry Molaison famously learned to trace the outline of a star while looking at his hand in a mirror (a surprisingly difficult task). His performance improved day after day — demonstrating procedural learning — even though he had no memory of ever having practiced the task before. Each session, he was surprised to be told he had done this many times.

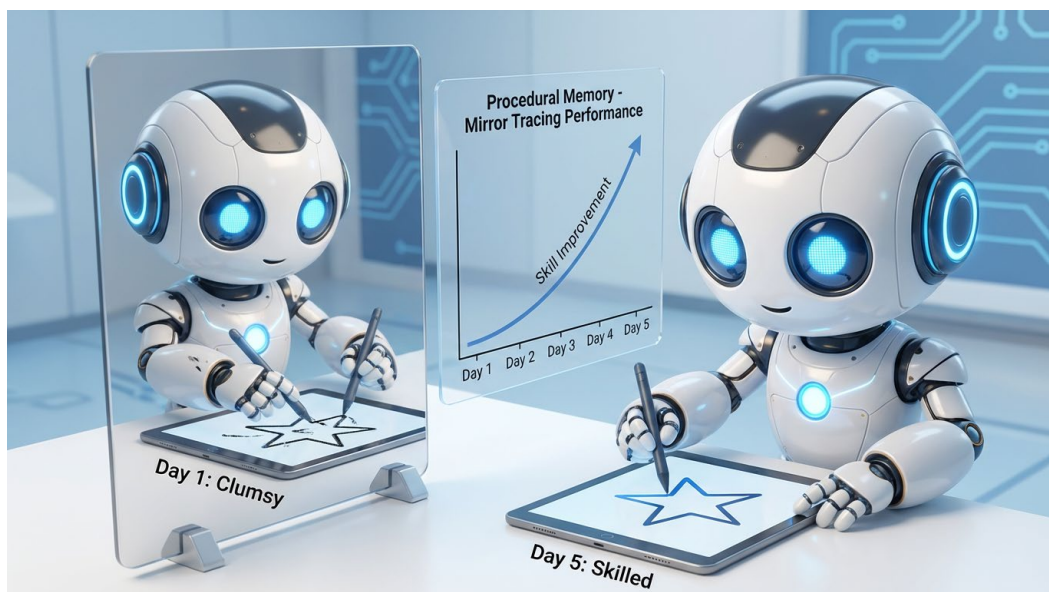
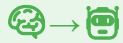


Figure 3: Henry Molaison's mirror tracing performance improved across days, despite no episodic memory of practice sessions.



AI CONNECTION: Procedural Memory in AI

In AI, procedural memory maps to learned policies in reinforcement learning. An RL agent's policy — the mapping from states to actions — is essentially a procedural skill. It tells the agent 'what to do' without explicit reasoning, just as your procedural memory tells your fingers where to go on a keyboard.

The analogy extends further: Just as procedural memory is built through repeated practice (and is slow to form), RL policies are trained through thousands or millions of episodes of trial and error.

Robotics makes this connection explicit: teaching a robot arm to grasp objects through reinforcement learning is building procedural memory in silicon. The robot cannot 'explain' its grasping strategy any more than you can explain how you balance on a bicycle.

Motor cortex → robot controller. Basal ganglia → action selection network. Cerebellum → forward model for predicting movement outcomes.

2.5 Putting It All Together — Four Systems, Four AI Parallels

Memory System	What It Stores	Key Brain Region	AI Parallel	Conscious?
Working	Current task info (temporary)	Prefrontal cortex	Context window, RNN hidden state	Yes
Episodic	Personal experiences (what, where, when)	Hippocampus	RAG, replay buffers, episodic control	Yes
Semantic	Facts and concepts (context-free)	Cortex (distributed)	Trained model weights	Yes
Procedural	Skills and habits (motor, cognitive)	Basal ganglia, cerebellum	RL policies, robot controllers	No

⚠ MISCONCEPTION ALERT

Common misconception: 'Memory is like a video recorder that stores everything perfectly.'

Reality: Memory is RECONSTRUCTIVE. Every time you recall an episode, your brain actively rebuilds it from fragments — and it can change in the process. Memories are not files stored on a hard drive; they are patterns reactivated across neural networks. This is why eyewitness testimony is notoriously unreliable, and why you might 'remember' events that never happened.

AI systems that store data digitally have perfect recall but lack the flexibility of reconstruction. Biological reconstruction enables generalization (you can imagine variations of past experiences) but introduces errors. There is a fundamental tradeoff between fidelity and flexibility.

? KNOWLEDGE CHECK

Test your understanding before moving on:

1. Name the four major memory systems and give one example of each from your daily life.
2. Which memory system was most affected by Henry Molaison's surgery? Which system remained intact, allowing him to learn mirror tracing?
3. A large language model's trained weights are most analogous to which type of biological memory? Why?
4. Why is the context window in a language model a limited analogy for working memory? What biological feature does it lack?
5. If someone has damage to their basal ganglia, which type of memory would be most affected?

Answers are discussed in the following sections — but try to answer from memory first!

Section 3

The Hippocampus — The Brain's Memory Engine

We introduced the hippocampus in Module 03 and touched on it in the previous section. Now we go deeper. The hippocampus is arguably the most important brain structure for understanding memory — and its computational principles have directly inspired some of the most important ideas in modern AI, including experience replay, memory consolidation, and complementary learning systems.

3.1 Anatomy and Circuitry

The hippocampus is a curved, elongated structure located in the medial temporal lobe, one in each hemisphere. Its name comes from the Greek for 'seahorse' because of its shape. It receives input from virtually all sensory cortices — visual, auditory, somatosensory — as well as from the prefrontal cortex and the amygdala. This makes it a convergence zone where information from different brain regions comes together.

The internal circuitry of the hippocampus follows a distinctive loop called the trisynaptic circuit. Information flows from the entorhinal cortex to the dentate gyrus, then to area CA3, then to area CA1, and back out to the entorhinal cortex. Each of these stages performs a different computational function, and understanding them reveals why the hippocampus is such an effective memory system.

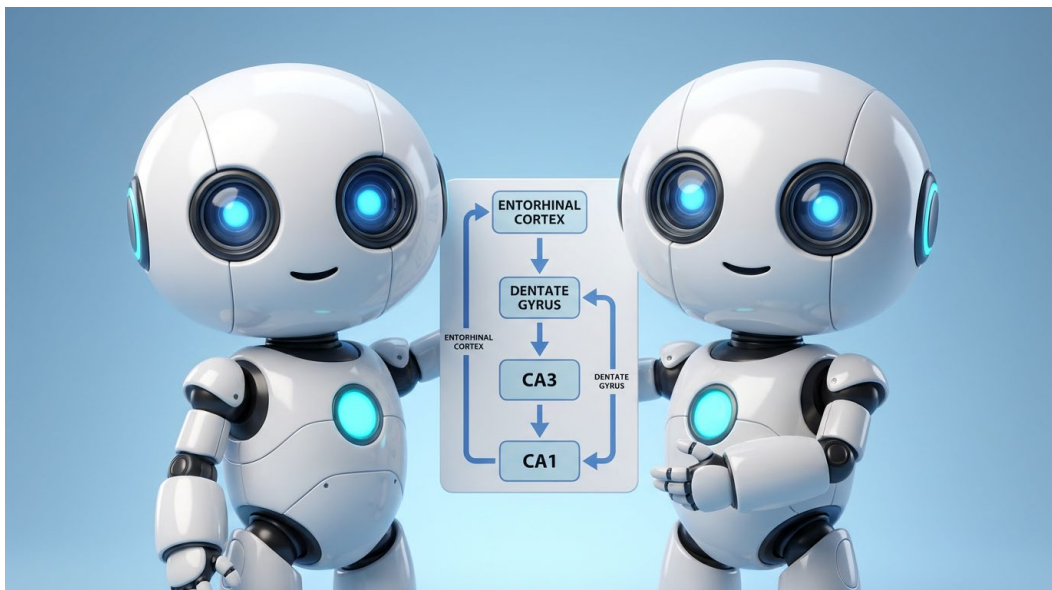


Figure 4: The hippocampal trisynaptic circuit — Entorhinal Cortex → Dentate Gyrus → CA3 → CA1 → back to Entorhinal Cortex.

The dentate gyrus is thought to perform pattern separation — taking similar inputs and making their representations more distinct. If two experiences are similar (say, parking your car in the same lot on two different days), the dentate gyrus helps create distinct memory traces so you do not confuse them. Area CA3 is thought to perform pattern completion — given a partial cue, it can reactivate an entire memory. If you smell a specific perfume and suddenly remember a complete scene from years ago, that is CA3 pattern completion at work. Area CA1 then compares the retrieved memory with current input, essentially asking: is this new, or have I seen this before?



AI CONNECTION: Pattern Separation and Completion

Pattern separation (dentate gyrus) is analogous to hashing in computer science — mapping similar inputs to distinct codes so they can be stored and retrieved without interference.

Pattern completion (CA3) is analogous to autoassociative memory or content-addressable memory — retrieving a full pattern from a partial cue. Hopfield networks (which John Hopfield won the Nobel Prize for in 2024) implement exactly this computation.

Modern AI systems typically lack explicit pattern separation. This contributes to 'catastrophic forgetting' — when training on new data overwrites old memories. Adding dentate gyrus-inspired mechanisms is an active area of research.

3.2 Memory Consolidation — Why You Need Sleep to Learn

The hippocampus does not store memories permanently. Instead, it serves as a temporary holding area — a fast-learning system that quickly encodes new experiences and then gradually transfers them to the cortex for long-term storage. This transfer process is called memory consolidation, and it happens primarily during sleep.

Here is how it works. During waking experience, the hippocampus rapidly encodes events — binding together the sights, sounds, emotions, and spatial context into a coherent episode. That night, during slow-wave sleep (deep sleep), the hippocampus spontaneously replays these recent experiences. Neurons fire in the same patterns they fired during the original experience, but compressed in time — what took minutes in real

life is replayed in seconds. This replay is thought to gradually strengthen the cortical connections that represent the memory, allowing the cortex to eventually store the memory independently of the hippocampus.

This is why sleep deprivation is so devastating for learning. Without adequate sleep, the hippocampus does not get the opportunity to replay and consolidate. Memories remain fragile and hippocampus-dependent, and the hippocampus fills up — it has limited capacity. Studies have shown that students who sleep after studying retain significantly more information than those who stay awake for the same period.



Figure 5: Memory consolidation during sleep — hippocampal replay transfers memories to cortical long-term storage.

KEY TAKEAWAY: The Two-System Architecture

The brain uses a TWO-SYSTEM memory architecture:

System 1 — Hippocampus: Fast learning, limited capacity, temporary storage. Quickly encodes new experiences but can only hold so much.

System 2 — Cortex: Slow learning, vast capacity, permanent storage. Gradually absorbs knowledge from the hippocampus through consolidation.

This solves a fundamental problem: if the cortex learned too fast, new information would overwrite old knowledge (catastrophic forgetting). If it learned too slowly, it could not keep up with new experiences. The hippocampus acts as a buffer — fast-learning on the front end, slow consolidation on the back end.

This two-system architecture is one of the most important ideas in both neuroscience and AI.

3.3 Experience Replay — From Hippocampus to DeepMind

In 2013, a small London-based startup called DeepMind published a paper that would change AI history. They trained a neural network to play Atari video games at superhuman levels — a system called DQN (Deep Q-Network). One of the key innovations that made DQN work was experience replay, and it was directly inspired by hippocampal replay during sleep.

The problem DeepMind faced was this: when a reinforcement learning agent learns from experiences as they happen (online learning), consecutive experiences are highly correlated — the next game frame looks very similar to the current one. This correlation destabilizes training because the neural network keeps adjusting to very similar examples, leading to oscillation and poor learning.

The solution was borrowed directly from the hippocampus. Instead of learning from experiences immediately, the agent stores them in a replay buffer — a large memory bank of past experiences. During training, the agent randomly samples from this buffer, breaking the correlations between consecutive experiences. This is exactly what the hippocampus does during sleep: it replays past experiences in a shuffled, compressed fashion, allowing the cortex to learn stable representations.

The parallel was not accidental. The DeepMind team, which included neuroscientists like Demis Hassabis (who had studied hippocampal memory during his PhD), explicitly drew on the neuroscience of memory consolidation. The DQN paper cited the biological inspiration, and subsequent work has only strengthened the connection.

Feature	Hippocampal Replay	DQN Experience Replay
When it happens	During sleep (slow-wave)	During training (between actions)
What is replayed	Recent waking experiences	Recent agent experiences (state, action, reward, next state)
Replay order	Compressed, often reversed or shuffled	Random sampling from buffer
Purpose	Consolidate to cortex, prevent forgetting	Stabilize training, break correlations
Capacity	Limited (hippocampus fills up)	Limited (buffer has fixed size)
Priority	Emotionally significant events replayed more	Prioritized replay weights by surprise (TD error)



THINK ABOUT IT

DeepMind later introduced Prioritized Experience Replay (2016), where experiences with higher surprise (larger prediction error) are replayed more often.

The brain likely does this too! Emotionally significant events (high amygdala activation) and surprising events (high dopamine prediction error) are consolidated more strongly. Both systems prioritize important memories over routine ones.

Question for discussion: Is this convergence — biology and AI arriving at the same solution — evidence that experience replay is a fundamental principle of efficient learning? Or just a coincidence?

3.4 Place Cells, Grid Cells, and Spatial Memory

Beyond episodic memory, the hippocampus plays a critical role in spatial navigation. In 1971, John O'Keefe discovered that specific hippocampal neurons — called place cells — fire when an animal is in a particular location in its environment. Each place cell has a 'place field' — a specific region of space where that neuron is most active. Together, the population of place cells creates an internal map of the environment.

In 2005, May-Britt Moser and Edvard Moser discovered grid cells in the entorhinal cortex (which provides input to the hippocampus). Grid cells fire in a remarkable hexagonal pattern as an animal moves through space, creating a coordinate system that measures

distance and direction. O'Keefe and the Mosers shared the Nobel Prize in Physiology or Medicine in 2014 for these discoveries.

The relationship between place cells and grid cells suggests that the brain builds a structured, geometric representation of space — a cognitive map. This is more than just a biological curiosity. In 2018, DeepMind showed that when they trained an artificial neural network to navigate, it spontaneously developed grid cell-like representations. The network was not designed to produce hexagonal firing patterns — they emerged naturally as the most efficient solution to the navigation problem. This is a powerful example of convergent evolution between biological and artificial systems.

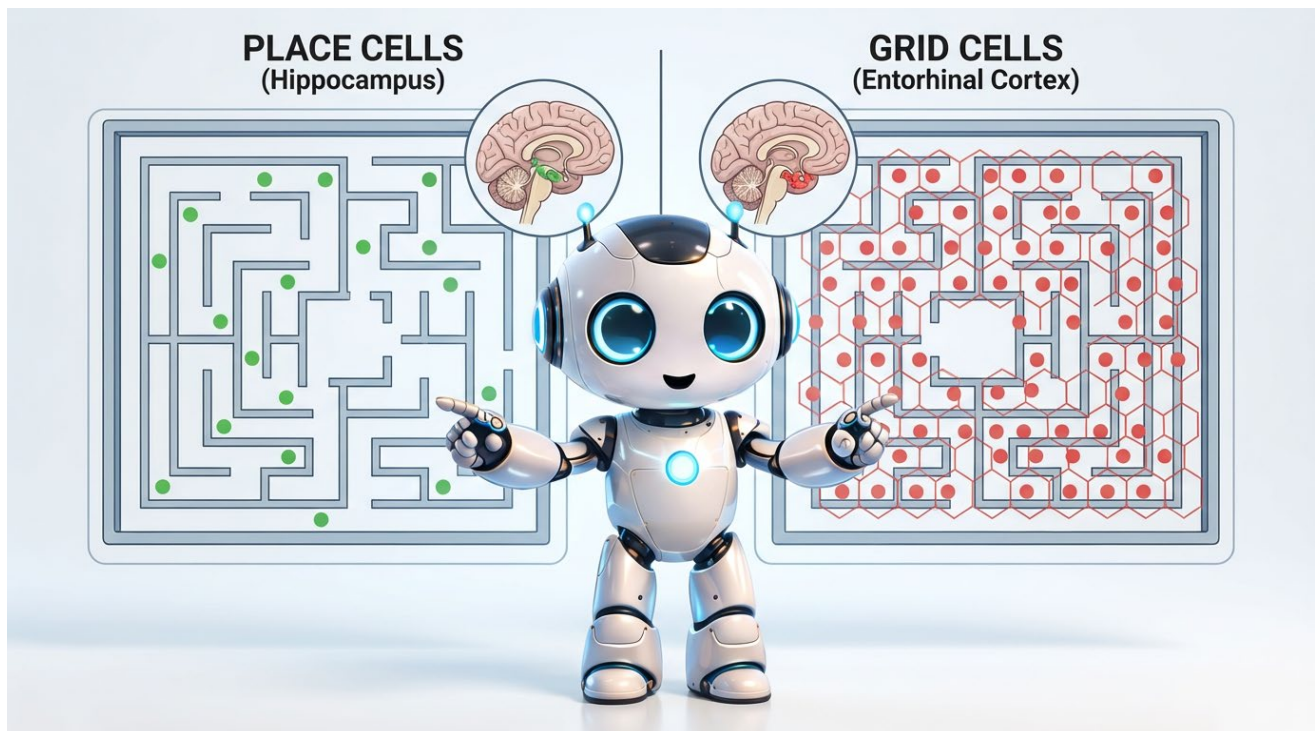
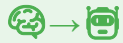


Figure 6: Place cells (hippocampus) fire at specific locations; grid cells (entorhinal cortex) fire in hexagonal patterns.



AI CONNECTION: Grid Cells in Artificial Agents

Banino et al. (2018, Nature) trained a recurrent neural network to navigate virtual environments using only velocity information. The network spontaneously developed grid cell-like units — hexagonal firing patterns identical to those found in the entorhinal cortex.

This suggests that grid cell representations are not a biological quirk but a mathematically optimal solution for spatial navigation. When AI faces the same problem the brain faces, it arrives at the same solution.

Applications: autonomous vehicle navigation, robot mapping (SLAM), and spatial reasoning in AI systems.

? KNOWLEDGE CHECK

Test your understanding before moving on:

1. What is the trisynaptic circuit? Name the three stations and the computational function each performs.
2. Explain why the brain uses a two-system architecture (hippocampus + cortex) rather than a single memory system.
3. How did hippocampal replay during sleep inspire DeepMind's experience replay in DQN?
4. What are place cells and grid cells? Why did the 2018 DeepMind result (artificial grid cells emerging in navigation networks) excite neuroscientists?
5. A student pulls an all-nighter before an exam instead of sleeping. Based on what you know about consolidation, what is likely to happen to their learning?

Answers are discussed in the following sections — but try to answer from memory first!

Section 4

Complementary Learning Systems — The Brain's Elegant Solution to Catastrophic Forgetting

We have described the two-system architecture of memory — the fast-learning hippocampus and the slow-learning cortex. In 1995, James McClelland, Bruce McNaughton, and Randall O'Reilly published one of the most influential papers in computational neuroscience: the Complementary Learning Systems (CLS) theory. This theory explains why the brain needs two learning systems and what goes wrong if you try to use just one.

The core problem is called catastrophic forgetting (also known as catastrophic interference). Imagine a standard artificial neural network that has been trained to recognize cats. Now you train it on a new task — recognizing dogs. If you simply train it on dog images, the weight changes needed for dogs will overwrite the weight patterns that encoded cats. When you test the network on cats again, performance has collapsed. The network has 'forgotten' cats by learning dogs.

This does not happen to you. You can learn to recognize dogs without forgetting what cats look like. The CLS theory explains how: your hippocampus quickly encodes each new experience without disturbing cortical knowledge, and then slowly — during sleep — it replays these experiences to the cortex, interleaving new knowledge with reactivated old knowledge. The cortex learns gradually from this interleaved stream, avoiding catastrophic overwriting.

KEY TAKEAWAY: Why Two Systems?

Fast learner alone (hippocampus only): Can quickly learn new things but has limited capacity and is fragile.

Slow learner alone (cortex only): Has vast capacity and is stable, but cannot quickly learn new things without overwriting old knowledge.

Both together: The hippocampus handles rapid learning, then gradually trains the cortex through interleaved replay. New knowledge is integrated with old knowledge without catastrophic forgetting.

This is why sleep is essential for learning — consolidation requires time for the hippocampus to 'teach' the cortex.

The CLS framework has become enormously influential in AI, particularly in the field of continual learning (also called lifelong learning). Standard neural networks suffer badly from catastrophic forgetting — this is a real, practical problem for deploying AI systems that need to learn new tasks over time without forgetting old ones. Researchers have developed several strategies inspired by CLS theory:

Strategy	Biological Inspiration	How It Works in AI
Experience replay	Hippocampal replay during sleep	Store past examples, mix with new data during training
Elastic Weight Consolidation (EWC)	Synaptic consolidation (important synapses resist change)	Protect important weights from changing when learning new tasks
Progressive neural networks	New cortical columns for new skills	Add new network modules for new tasks, keep old ones frozen
Generative replay	Hippocampal replay generates reconstructed memories	Train a generative model to 'dream up' old examples to mix with new data

THINK ABOUT IT

Catastrophic forgetting is not just an academic problem. Imagine a self-driving car trained in Phoenix (dry, flat roads) that is redeployed to Seattle (wet, hilly roads). If you fine-tune it on Seattle data, you do not want it to forget how to drive in Phoenix. Complementary Learning Systems theory provides a principled framework for solving this problem.

Section 5

How the Brain Learns — The Cellular and Chemical Machinery

We have explored where memories live and how they move between brain systems. Now we turn to a more fundamental question: how does the brain actually change when it learns something? What physically happens inside your head when you study a new concept, practice a new skill, or experience something for the first time?

The short answer is that learning changes the strength of connections between neurons. But the details of how this happens — the specific mechanisms the brain uses, the chemical signals that control when learning occurs, and the different speeds at which different memory systems learn — are fascinating, and they reveal principles that AI researchers are only beginning to adopt.

5.1 Long-Term Potentiation — The Volume Knob Between Neurons

In 1973, Tim Bliss and Terje Lømo made one of the most important discoveries in all of neuroscience. They were studying neurons in the hippocampus of a rabbit, and they found something remarkable: if they stimulated a neural pathway with a burst of rapid, repeated electrical pulses (like a quick drumroll), the connection between those neurons became stronger — and stayed stronger for hours, days, or even weeks. They called this phenomenon Long-Term Potentiation, or LTP.

Think of it this way. Imagine two friends who talk on the phone occasionally. One day, they have an intense, rapid-fire conversation — lots of back and forth in a short period. After that conversation, the connection between them is stronger. They call each other more easily, they respond more quickly, and the signal between them is clearer. That is essentially what LTP does between neurons.

Before LTP was discovered, scientists knew that learning must somehow change the brain, but nobody had caught the brain in the act of changing. LTP was the first direct evidence that neural connections can be strengthened by experience — the cellular basis of learning.

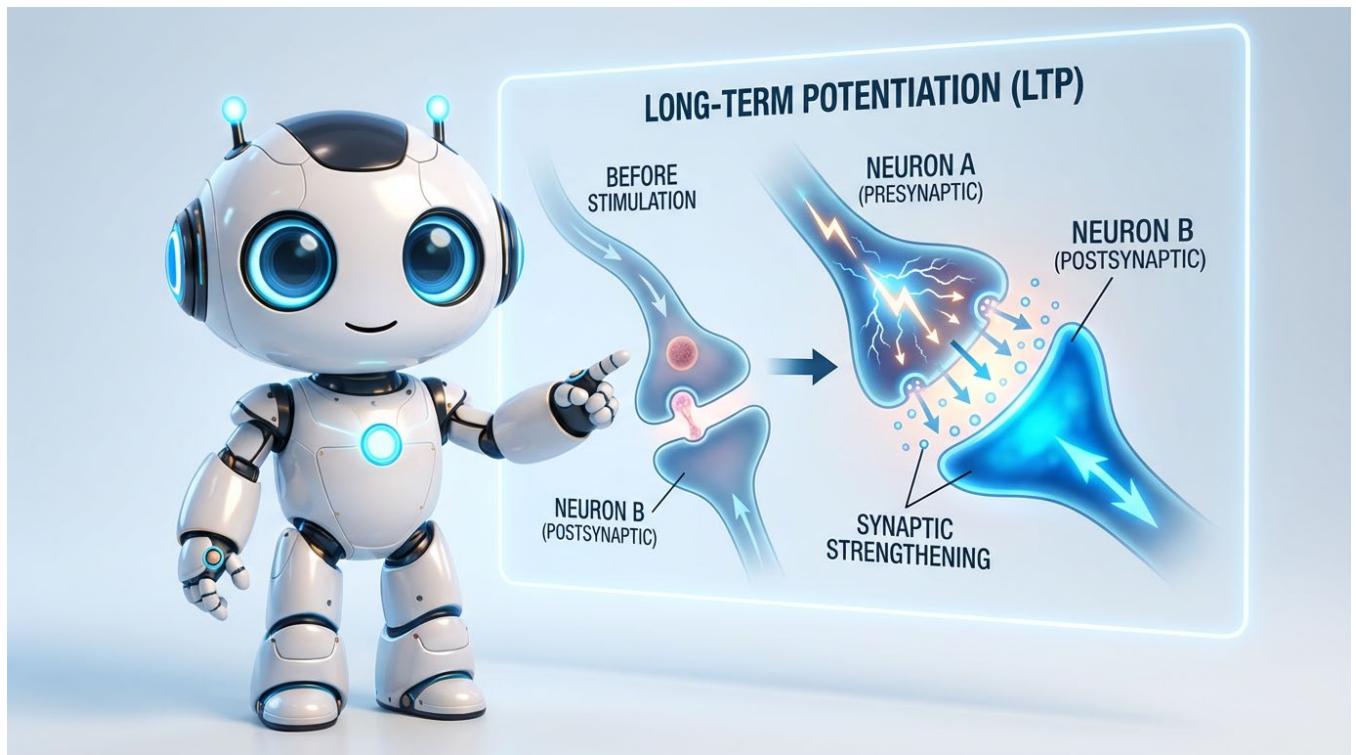


Figure 7: Long-Term Potentiation (LTP) — a burst of stimulation permanently strengthens the connection between two neurons.

Here is how LTP works at the cellular level, explained simply. When neuron A sends a signal to neuron B, it releases a chemical messenger (a neurotransmitter called glutamate) into the gap between them (the synapse). Neuron B has receivers (receptors) that detect this glutamate. Under normal conditions, only some of these receptors are active — think of it as neuron B listening with one ear.

During LTP, something special happens. When the stimulation is strong and repeated, a particular type of receptor called the NMDA receptor opens up. The NMDA receptor is special because it acts as a coincidence detector — it only opens when two things happen simultaneously: the presynaptic neuron is sending a signal AND the postsynaptic neuron is already somewhat active. When the NMDA receptor opens, it allows calcium ions to flood into neuron B. This calcium surge triggers a cascade of changes: more receptors are inserted into the membrane (now neuron B is listening with both ears), the synapse physically grows larger, and new proteins are made to maintain these changes long-term.

**KEY TAKEAWAY: LTP Is Hebbian Learning Made Real**

Remember Hebb's rule from Module 04: 'neurons that fire together wire together'?

LTP is the biological mechanism that makes Hebb's rule work. The NMDA receptor is the coincidence detector — it requires BOTH neurons to be active simultaneously (Hebb's condition). When that condition is met, the connection is strengthened (Hebb's result).

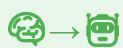
This is not just a theoretical principle — it is a physical, chemical process happening in your brain right now as you read this booklet and form new memories.

5.2 Long-Term Depression — The Brain's Eraser

If the brain could only strengthen connections, it would quickly become saturated — every connection at maximum strength, with no way to distinguish important patterns from noise. The brain needs an eraser as much as it needs a pencil. That eraser is Long-Term Depression, or LTD.

LTD is the opposite of LTP: it weakens the connection between neurons. It occurs when stimulation is slow and low-frequency — the neural equivalent of a lazy, infrequent conversation rather than an intense exchange. While LTP says 'this connection matters, make it stronger,' LTD says 'this connection is not being used much, let it fade.'

Together, LTP and LTD give the brain a bidirectional volume knob for every single connection. Turn it up (LTP) for connections that are useful. Turn it down (LTD) for connections that are not. This balance is essential for learning — it allows the brain to sharpen important signals while suppressing irrelevant ones, just as a sculptor removes material to reveal a shape.

**AI CONNECTION: LTP/LTD and Weight Updates in Neural Networks**

LTP and LTD together are the biological version of weight updates in artificial neural networks:

LTP (strengthening) = increasing a weight value during training

LTD (weakening) = decreasing a weight value during training

In backpropagation, weight changes can be positive or negative based on the error gradient. In the brain, LTP and LTD provide the same bidirectional adjustment using completely different biological mechanisms.

One key difference: in AI, weight changes are computed by an algorithm (backprop). In the brain, they emerge from local chemical interactions at each synapse. No central controller tells each synapse how much to change — the synapse figures it out from its own local signals.

Another difference: the brain's weight changes (LTP/LTD) have built-in stability mechanisms — there are limits to how strong or weak a connection can get. Artificial networks sometimes need explicit techniques (like weight clipping or batch normalization) to achieve similar stability.



MATH MOMENT (Optional): LTP/LTD as a Learning Rule

We can express LTP and LTD as a simple learning rule:

If presynaptic and postsynaptic neurons fire together (high correlation):

$\Delta w > 0 \rightarrow$ connection strengthened (LTP)

If presynaptic neuron fires but postsynaptic does not (low correlation):

$\Delta w < 0 \rightarrow$ connection weakened (LTD)

This is exactly the STDP rule from Module 04, but described at the cellular/chemical level rather than the spike-timing level. STDP is the precise timing version of the general LTP/LTD principle.

Compare to a simple artificial weight update: $\Delta w = \eta \cdot (\text{target} - \text{output}) \cdot \text{input}$

Both are changing connection strength based on activity patterns. The brain does it with chemistry; AI does it with calculus.

5.3 The Chemical Gatekeepers — Neuromodulators That Control When You Learn

LTP and LTD explain how connections change. But the brain also needs to control when connections change. You do not want every single experience to permanently rewire your brain — most of what happens in a day is routine and forgettable. The brain needs a way to say 'this is important, learn it now' or 'this is routine, ignore it.'

This gating function is performed by neuromodulators — chemical signals that do not directly cause neurons to fire but instead change how responsive neurons are to other inputs. Think of neuromodulators as the thermostat in a building: they do not generate heat, but they control when the heating system turns on. Three neuromodulators are especially important for learning:

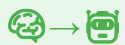
Dopamine — The 'That Was Surprising' Signal. We met dopamine in Module 03 when we discussed the Schultz et al. (1997) discovery: dopamine neurons fire when something is *better than expected* (positive surprise), pause when something is *worse than expected* (negative surprise), and do nothing when things go *exactly as expected*. This is the reward prediction error — the exact same signal used in temporal difference learning in AI. In terms of learning, dopamine acts as a gate: when dopamine is released, it enhances LTP at recently active synapses. Translation: when something surprising and good happens, the brain strengthens whatever connections were active around that time. This is how we learn from rewards.

Acetylcholine — The 'Pay Attention' Signal. Acetylcholine is released by a brain region called the basal forebrain, and its levels increase when you are alert, focused, and attending to something. Acetylcholine enhances the brain's sensitivity to incoming sensory information while suppressing internal noise and old memories. Think of it as turning up the microphone for new input while turning down the volume on old recordings. When acetylcholine levels are high, LTP is enhanced — you learn new information more effectively. This is why paying attention matters so much for learning: attention literally changes the chemistry of your synapses. It also explains why diseases

that destroy acetylcholine neurons (like Alzheimer's disease) devastate the ability to form new memories.

Norepinephrine — The 'Something Important Is Happening' Signal.

Norepinephrine (also called noradrenaline) is released by a brain region called the locus coeruleus during states of arousal, stress, or novelty. It broadly increases alertness and enhances learning across multiple brain areas. Norepinephrine is why you remember where you were during a shocking news event but not what you had for lunch on a random Tuesday — emotional arousal releases norepinephrine, which enhances memory formation. It also plays a role in the explore-versus-exploit tradeoff: high norepinephrine promotes exploration (trying new things), while low norepinephrine promotes exploitation (sticking with what works).



AI CONNECTION: Neuromodulators and AI Learning Rates

Neuromodulators control the brain's effective 'learning rate' — how much a single experience changes the system. AI uses a fixed or scheduled learning rate (η). The brain's learning rate is dynamic, context-dependent, and controlled by multiple chemical systems simultaneously.

Dopamine → Reward-based learning rate: In deep RL, the learning rate is fixed. In the brain, dopamine scales the learning rate based on surprise. This is more like adaptive learning rate methods (Adam, RMSprop) but tied to motivational significance.

Acetylcholine → Attention-gated learning: Some AI architectures use attention to gate what gets processed, but few use attention to gate what gets **LEARNED**. This is an open frontier.

Norepinephrine → Exploration vs. exploitation: RL agents use ϵ -greedy or temperature parameters to balance exploration and exploitation. The brain uses norepinephrine. The Aston-Jones & Cohen (2005) model proposes that norepinephrine controls a 'neural gain' parameter — essentially tuning how exploratory or focused the entire brain is.

Key gap: AI systems typically use one learning rate for all parameters. The brain uses different neuromodulators to create different learning rates in different brain regions for different types of information — simultaneously.

5.4 Different Memory Systems Learn at Different Speeds

One of the most important and overlooked aspects of biological learning is that different memory systems operate on completely different timescales. This is not a design flaw — it is a feature. Each memory system is optimized for a different type of learning problem.

Memory System	Learning Speed	How Many Exposures?	Everyday Example
Episodic (hippocampus)	Extremely fast — one shot	Just ONE experience	You remember the restaurant where you got food poisoning after a single visit
Semantic (cortex)	Slow — gradual extraction	Many exposures over weeks/months	You slowly build up knowledge of world history from many classes and readings
Procedural (basal ganglia)	Very slow — repetition-based	Thousands of practice trials	It takes months of daily practice to learn to play piano fluently
Working memory (PFC)	Instant — but temporary	Zero (it is active maintenance)	You hold a phone number in mind for seconds while dialing

This range of learning speeds — from one-shot episodic learning to the thousands of repetitions needed for procedural skills — is striking when compared to AI. Most artificial neural networks have a single learning rate that applies uniformly to all parameters. The brain, by contrast, has evolved specialized systems with learning rates that span several orders of magnitude, each tuned to a different type of information.



THINK ABOUT IT

One-shot learning is one of the most impressive things the brain does. You eat a new food, get sick, and NEVER eat it again — one trial, permanent learning. AI systems typically need hundreds or thousands of examples to learn a single concept reliably.

Few-shot learning in AI (like GPT-4 learning from a few examples in the prompt) is a step toward this, but it works through the context window (working memory), not through permanent weight changes. The brain's one-shot learning actually rewires the hippocampus permanently in a single experience.

Question: Why might it be dangerous for an AI system to learn permanently from a single example, the way the hippocampus does? (Hint: think about adversarial examples and hallucinations.)

5.5 Sleep Stages and Learning — Not All Sleep Is Equal

In Section 3 we discussed how the hippocampus replays experiences during sleep to consolidate them into the cortex. But sleep is not a uniform state — it consists of distinct stages, and different stages play different roles in different types of learning. Understanding this reveals another layer of sophistication in the brain's learning machinery.

Sleep alternates between two main states throughout the night: Non-REM sleep (which includes light sleep and deep slow-wave sleep) and REM sleep (Rapid Eye Movement sleep, when most dreaming occurs). A typical night involves four or five cycles through these stages, each cycle lasting about ninety minutes.

Sleep Stage	What Happens in the Brain	What Type of Learning It Helps	How It Works
Slow-Wave Sleep (deep Non-REM)	Hippocampal replay: slow oscillations replay the day's experiences to the cortex	Episodic and semantic memory: facts, events, textbook material	Hippocampus 'teaches' the cortex by replaying compressed episodes; cortex gradually builds stable representations
REM Sleep (dreaming)	Cortical networks are active; hippocampus is relatively quiet; brain replays and recombines memories creatively	Procedural memory: motor skills, emotional processing, creative insight	Motor circuits rehearse recently practiced movements; emotional memories are processed and integrated; novel combinations of memories can produce creative insights

This division of labor during sleep has practical implications. If you are studying for an exam (episodic and semantic learning), getting enough deep slow-wave sleep is critical — this tends to occur more in the first half of the night. If you are practicing a musical instrument or a sport (procedural learning), REM sleep matters more — and REM sleep tends to increase in the second half of the night. Students who go to bed early and wake up early may get more slow-wave consolidation; those who sleep late may get more REM consolidation.

⚠ MISCONCEPTION ALERT

Common misconception: 'I can learn just as well if I skip sleep and study more hours.'

Reality: Sleep is not wasted time — it is when learning actually solidifies. During sleep, the brain is actively replaying, reorganizing, and strengthening what you studied. Multiple studies show that a good night's sleep after studying produces better performance than extra hours of studying without sleep.

Robert Stickgold's research at Harvard showed that people who slept after learning a visual discrimination task improved their performance by 20% the next day — even without additional practice. People who stayed awake showed no improvement at all.

🧠 → 🤖 AI CONNECTION: Sleep Stages and AI Training Schedules

AI systems do not sleep, but the concept of different consolidation phases has parallels:

Slow-wave sleep (hippocampal replay) → Experience replay in RL: both involve replaying stored experiences to a slow-learning system. Some researchers have experimented with 'sleep phases' in AI training where the network alternates between learning from new data and replaying old data.

REM sleep (creative recombination) → Generative replay and data augmentation: training on recombined or generated variations of past experiences, similar to how REM sleep combines memory fragments in novel ways.

A 2022 paper by Tadros et al. proposed 'Sleep Replay Consolidation' (SRC) for continual learning in AI — directly modeling the slow-wave/REM cycle. Networks that 'slept' between tasks showed significantly less catastrophic forgetting.

The broader lesson: biological learning is not just about the learning events themselves — it is equally about what happens BETWEEN learning events. AI has mostly focused on optimizing the learning step; the brain teaches us that the consolidation step matters just as much.

? KNOWLEDGE CHECK

Test your understanding before moving on:

1. What is LTP? Use the 'phone call' analogy to explain it to a friend who has never taken a neuroscience class.
2. Why does the brain need both LTP (strengthening) and LTD (weakening)? What would happen if it only had LTP?
3. Name the three major neuromodulators discussed in this section. For each one, describe in one sentence what it signals and how it affects learning.
4. Why is one-shot learning in the hippocampus both a strength and a potential risk? How does this compare to how AI systems learn?
5. A student practices piano for two hours, then studies history for two hours, then goes to sleep. Which sleep stage will help consolidate the piano practice? Which will help consolidate the history facts?

Answers are discussed in the following sections — but try to answer from memory first!

PART 2

How AI Remembers — And What It Gets Wrong

Computational Parallels — With Optional Math Moments

How AI Remembers — And What It Gets Wrong



Section 6

AI Memory Architectures — Building Remembering Machines

Now that we understand how the brain stores, consolidates, and retrieves memories, let us examine how AI systems have approached the memory problem. Some of these approaches were directly inspired by neuroscience; others were independently invented but converge on similar principles. In each case, we will identify what the AI system does well, where it falls short compared to biology, and what the open challenges are.

6.1 Recurrent Neural Networks and LSTMs — The First AI Memory

Standard feedforward neural networks have no memory at all — they process each input independently, with no awareness of what came before. Recurrent neural networks (RNNs), introduced in the 1980s, solved this by adding a feedback loop: the output of the network at one time step is fed back as input at the next step. This creates a hidden state — a vector that accumulates information over time, acting as a form of working memory.

In principle, RNNs can remember indefinitely. In practice, they suffer from the vanishing gradient problem: information from early time steps fades exponentially as it passes through many layers of processing. An RNN trying to remember something from a hundred time steps ago is like trying to hear a whisper across a football stadium — the signal has degraded to nothing.

Long Short-Term Memory networks (LSTMs), invented by Sepp Hochreiter and Jürgen Schmidhuber in 1997, addressed this with a gating mechanism. An LSTM cell has three gates — an input gate (what new information to store), a forget gate (what old information to discard), and an output gate (what stored information to use right now). The key innovation is the cell state — a highway that runs through time with minimal modification, allowing information to persist for hundreds of time steps without degradation.

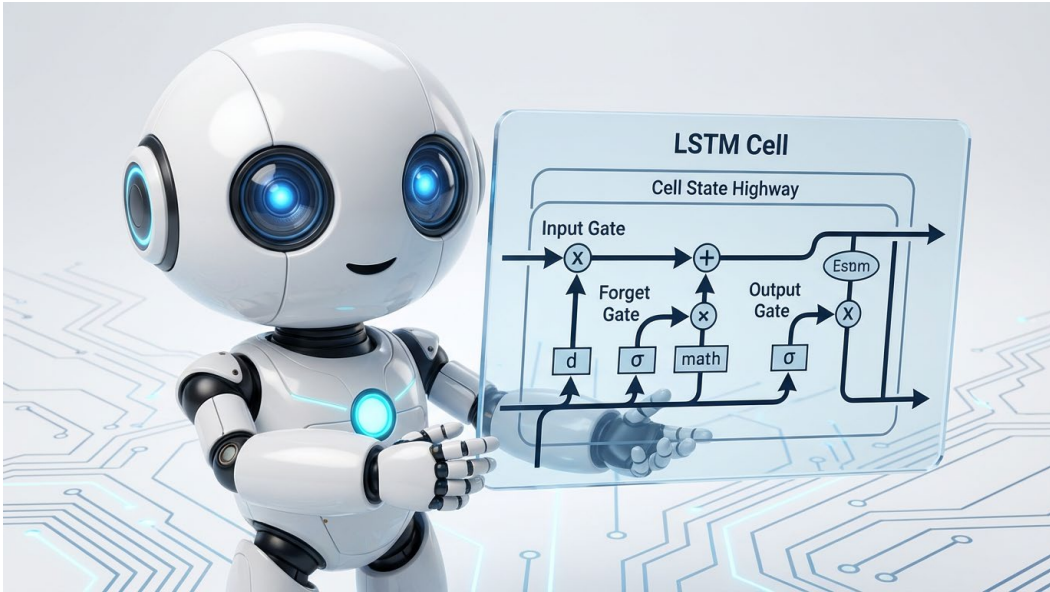


Figure 8: LSTM cell architecture — input gate, forget gate, output gate, and the cell state highway.



AI CONNECTION: LSTM Gates and Biological Gating

The LSTM's gating mechanism has a biological parallel in the thalamus — the brain's relay station that gates which information reaches the cortex. The prefrontal cortex controls the thalamic gate, deciding what information to let through based on task demands.

The forget gate is particularly interesting biologically. The brain actively forgets — it has mechanisms (like synaptic depression and protein degradation) that weaken memories. This is not a bug; it is essential for efficient operation. A brain that could not forget would be overwhelmed by irrelevant detail.

The cell state highway in LSTMs mirrors the concept of 'protected information channels' in the brain — certain neural pathways that maintain stable representations despite ongoing processing in other circuits.



MATH MOMENT (Optional): The LSTM Gating Equations

If you are curious about the math behind LSTM gates, here is what each gate computes:

$$\text{Forget gate: } f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$\text{Input gate: } i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\text{Candidate: } \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

Cell state: $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$

Output gate: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$

Hidden state: $h_t = o_t * \tanh(C_t)$

σ is the sigmoid function (outputs 0–1, acting as a 'gate').

The forget gate decides what percentage of the old cell state to keep.

The input gate decides how much of the new candidate to add.

The cell state C_t carries information through time — the protected highway.

You do NOT need to memorize these equations. The conceptual understanding of gates is what matters.

6.2 Transformers and Attention — Memory Through Retrieval

If you have taken ITAI 2376 (Deep Learning), you have already studied transformers and the attention mechanism. Here we examine them through a neuroscience lens — specifically, how attention in transformers relates to memory and attention in the brain.

The transformer architecture, introduced by Vaswani et al. in 2017 ('Attention Is All You Need'), replaced the sequential processing of RNNs with a parallel mechanism called self-attention. In self-attention, every element in a sequence can directly attend to every other element, regardless of distance. This eliminates the vanishing gradient problem entirely — a word at position 1 can directly influence a word at position 1000 with the same strength.

From a memory perspective, transformers implement a form of content-addressable memory. Each token in the context window generates a query ('what am I looking for?'), a key ('what do I contain?'), and a value ('what information do I provide?'). Attention scores are computed by matching queries to keys — essentially asking 'which other tokens are relevant to me?' This is remarkably similar to how the hippocampus retrieves memories: a partial cue (the query) activates stored patterns (key-value pairs) based on similarity.



AI CONNECTION: Attention and Biological Attention

The term 'attention' in AI was not chosen randomly — it was inspired by biological attention, the brain's ability to selectively focus on relevant information while ignoring irrelevant input.

Biological attention operates at multiple levels:

- Spatial attention: focusing on a specific location (controlled by parietal cortex)
- Feature attention: focusing on a specific property like color or motion (controlled by prefrontal-parietal circuits)
- Memory-guided attention: using past experience to guide what to attend to (hippocampal-prefrontal interaction)

Transformer attention most closely resembles memory-guided attention — using stored representations to decide what is relevant to the current context.

Key difference: Biological attention is **SELECTIVE** (focus on one or few things, ignore others). Transformer attention attends to **EVERYTHING**, just with different weights. This is more like a 'soft spotlight' compared to the brain's 'hard spotlight.'



MATH MOMENT (Optional): Self-Attention Computation

The core computation of self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(Q \cdot K^T / \sqrt{d_k}) \cdot V$$

Q (queries): 'what am I looking for?'

K (keys): 'what do I contain?'

V (values): 'what information do I provide?'

d_k : dimension of the key vectors (scaling factor to prevent huge dot products)

The softmax creates a probability distribution — a weighted combination of values based on query-key similarity. This is a differentiable version of nearest-neighbor lookup in memory.

Multi-head attention runs this computation multiple times in parallel with different learned projections — similar to how the brain processes different aspects of a stimulus in parallel streams.

6.3 Memory-Augmented Neural Networks — External Memory Banks

LSTMs and transformers have memory built into their architecture — hidden states and context windows, respectively. But both have fixed capacity. What if a neural network could read from and write to an external memory bank, like a person consulting a notebook?

This is exactly what memory-augmented neural networks (MANNs) do. The most famous examples are the Neural Turing Machine (NTM, Graves et al., 2014) and the Differentiable Neural Computer (DNC, Graves et al., 2016), both from DeepMind. These architectures separate the neural network (the 'controller') from the memory (the 'memory matrix'), just as a CPU is separate from RAM in a computer — and just as the hippocampus is architecturally distinct from the cortex in the brain.

The controller can read from memory (using attention-based retrieval, similar to hippocampal pattern completion), write to memory (encoding new information, similar to hippocampal encoding), and erase from memory (removing outdated information, similar to active forgetting). The memory matrix is accessed via attention mechanisms, making the entire system differentiable and trainable with backpropagation.

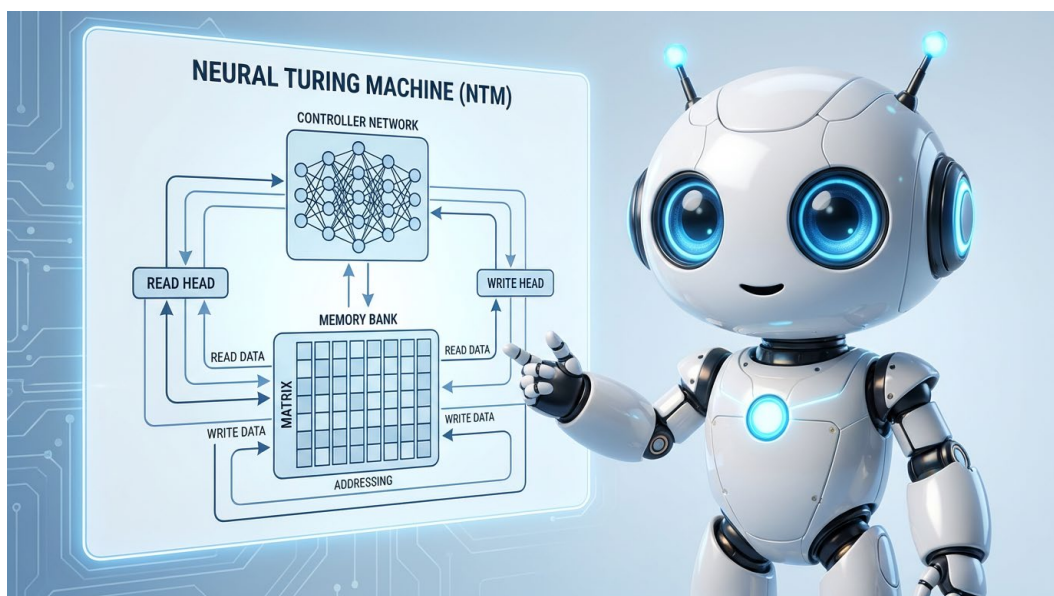


Figure 9: Neural Turing Machine architecture — a controller network with separate read/write heads accessing an external memory matrix.



AI CONNECTION: NTM/DNC and the Hippocampus

The parallel between MANNs and the hippocampal-cortical system is striking:

NTM controller = Prefrontal cortex (decides what to read, write, erase)

Memory matrix = Hippocampus (stores patterns, supports content-addressable retrieval)

Read head = Pattern completion (CA3, retrieving full memories from partial cues)

Write head = Encoding (hippocampal fast learning of new experiences)

Erase head = Active forgetting (synaptic depression, memory clearance)

MANNs can learn to copy sequences, sort lists, and navigate graphs — tasks that require explicit memory operations. However, they remain difficult to scale to very large memories.

6.4 Retrieval-Augmented Generation (RAG) — The Modern Approach

The most practical approach to AI memory today is Retrieval-Augmented Generation (RAG). Instead of trying to store everything in a neural network's parameters, RAG systems maintain a separate database of documents, facts, or past interactions. When the AI receives a query, it first searches this database for relevant information, retrieves the most pertinent passages, and then generates a response informed by both its parametric knowledge (model weights) and the retrieved information.

RAG maps elegantly onto the hippocampal-cortical distinction. The neural network's trained weights are like cortical semantic memory — general knowledge extracted from extensive training. The external database is like the hippocampus — a fast-updated store of specific experiences and facts that can be retrieved on demand. Just as the hippocampus provides specific episodic details to complement the cortex's general knowledge, the RAG database provides specific, up-to-date information to complement the model's general capabilities.

💡 WHY RAG MATTERS FOR YOUR CAREER

RAG is currently one of the most in-demand skills in AI engineering. Companies use RAG to build chatbots that can answer questions about their specific products, internal documents, and customer data — without retraining the entire language model.

If you understand the neuroscience behind RAG (hippocampal retrieval, content-addressable memory, pattern completion), you will have a deeper understanding of why it works and where it fails — giving you an edge over engineers who only know the technical implementation.

6.5 Comparing AI Memory Architectures

Architecture	Memory Type	Capacity	Biological Parallel	Best For
RNN/LSTM	Internal (hidden state)	Small, fixed	Working memory (PFC)	Short sequences
Transformer	Internal (context window)	Medium, fixed	Working memory + attention	Text, code, reasoning
NTM/DNC	External (memory matrix)	Medium, flexible	Hippocampal read/write	Algorithmic tasks
RAG	External (database)	Very large	Hippocampal-cortical retrieval	Knowledge-intensive QA
Experience replay	Buffer (replay memory)	Large, FIFO	Hippocampal replay in sleep	Reinforcement learning

? KNOWLEDGE CHECK

Test your understanding before moving on:

1. What problem do LSTMs solve that standard RNNs cannot? What mechanism do they use?
2. How is self-attention in transformers similar to memory retrieval in the hippocampus?
3. What is the key architectural innovation of Memory-Augmented Neural Networks like the Neural Turing Machine?
4. How does RAG mirror the hippocampal-cortical distinction in biological memory?
5. Which AI memory architecture would you use for a customer service chatbot that needs to reference specific company documents? Why?

Answers are discussed in the following sections — but try to answer from memory first!

Section 7

The Biological Plausibility Debate — Can the Brain Do Backpropagation?

Throughout this course, we have drawn parallels between biological and artificial learning systems. Now it is time to address the elephant in the room: how does the brain actually learn? And does it use anything like backpropagation — the algorithm that powers virtually all of modern deep learning?

This is not just an academic question. If the brain uses fundamentally different learning principles than backpropagation, then the brain may hold the key to learning algorithms that are more efficient, more flexible, and more robust than what we currently have. Understanding this debate is essential for anyone working at the intersection of neuroscience and AI.

7.1 How Backpropagation Works (Conceptual Review)

Backpropagation is the standard algorithm for training neural networks. Conceptually, it works in two phases. In the forward pass, input data flows through the network layer by layer, producing an output. This output is compared to the desired answer, producing an error signal. In the backward pass, this error signal flows backward through the network — from output to input — and each connection weight is adjusted in proportion to how much it contributed to the error. The key insight is the chain rule of calculus, which allows the error to be efficiently propagated backward through many layers.

Backpropagation is extraordinarily effective. It has powered every major advance in deep learning, from image recognition to language models to game-playing agents. But it has several properties that are biologically implausible — meaning the brain almost certainly does not implement it in this exact form.

7.2 Why Backprop Is Biologically Implausible

There are at least four major reasons why backpropagation, as implemented in artificial neural networks, is unlikely to occur in biological brains:

Problem 1 — The Weight Transport Problem. In backpropagation, the backward pass must use the exact same weights as the forward pass, but in reverse. In biological neurons, the forward connection from neuron A to neuron B is a physical synapse. For backpropagation to work biologically, there would need to be a separate backward connection from B to A carrying the exact same weight — but there is no known biological mechanism for maintaining such symmetric connections. The brain has feedback connections, but they do not carry copies of feedforward weights.

Problem 2 — Non-local Error Signals. In backpropagation, a neuron deep in the network adjusts its weights based on the error at the output layer — potentially many layers away. But biological neurons primarily have access to local information: their own activity, the activity of their immediate neighbors, and local neuromodulatory signals like dopamine. There is no known mechanism for precisely propagating error signals across many synaptic layers.

Problem 3 — Separate Forward and Backward Phases. Backpropagation requires two distinct phases — a forward pass to compute the output, and a backward pass to compute gradients. The brain does not appear to have separate computational phases. Neural activity is continuous, and neurons are simultaneously sending signals forward and receiving signals from later areas. The brain's processing is more like a continuous, bidirectional flow than a neat two-phase computation.

Problem 4 — Real-Valued Error Derivatives. Backpropagation computes precise real-valued derivatives. Biological neurons communicate with all-or-nothing spikes. While firing rates can encode analog values, the precision required for gradient computation seems far beyond what spiking neurons can achieve.

7.3 Hebbian Learning — The Brain's Learning Rule

If the brain does not use backpropagation, what does it use? The oldest and most established biological learning rule is Hebbian learning, proposed by Donald Hebb in 1949. Hebb's famous principle states: 'Neurons that fire together wire together.' When two connected neurons are active at the same time, the connection between them is strengthened. When they fire out of sync, the connection weakens.

We encountered a precise version of Hebbian learning in Module 04: Spike-Timing-Dependent Plasticity (STDP). In STDP, if a presynaptic neuron fires just before a postsynaptic neuron (suggesting it helped cause the postsynaptic firing), the connection is strengthened. If the presynaptic neuron fires just after the postsynaptic neuron (suggesting it was not involved in causing the firing), the connection is weakened. The direction and magnitude of the change depend on the precise timing difference between the two spikes.

Hebbian learning is entirely local — each synapse adjusts based only on the activity of its own pre- and postsynaptic neurons. It does not require a global error signal, weight transport, or separate computational phases. This makes it biologically plausible. However, pure Hebbian learning has significant limitations: it is unstable (weights can grow without bound), it cannot learn complex input-output mappings, and it has no concept of a task objective or error correction.

MATH MOMENT (Optional): Hebbian vs. Backprop Weight Updates

Hebbian update: $\Delta w_{ij} = \eta \cdot x_i \cdot x_j$

(Change in weight = learning rate $\times$ presynaptic activity $\times$ postsynaptic activity)

Purely local — only needs information at the synapse.

Backprop update: $\Delta w_{ij} = \eta \cdot \delta_j \cdot x_i$

(Change in weight = learning rate $\times$ error signal from output $\times$ presynaptic activity)

Non-local — requires error δ_j propagated from the output layer.

The crucial difference: Hebbian learning has no error signal. It strengthens correlations but does not know what the 'right answer' should be. Backprop explicitly uses the error to correct the network.

This is why Hebbian learning alone cannot train deep networks — it has no way to assign credit for errors to specific weights deep in the network (the 'credit assignment problem').

7.4 Bridging the Gap — Biologically Plausible Learning Algorithms

Researchers have spent decades trying to find learning algorithms that are both biologically plausible and powerful enough to train deep networks. Several promising approaches have emerged:

Feedback Alignment (Lillicrap et al., 2016): Instead of using the exact transpose of the forward weights for the backward pass (which requires weight transport), feedback alignment uses random, fixed feedback weights. Surprisingly, this works — the forward weights gradually align with the random feedback weights during training, and the network learns nearly as well as with standard backpropagation. This eliminates the weight transport problem while preserving the ability to train deep networks.

Predictive Coding (Rao & Ballard, 1999): We introduced predictive coding in Module 02 as a theory of brain function. It turns out that predictive coding networks — where each layer tries to predict the activity of the layer below it, and errors are propagated upward — can perform learning equivalent to backpropagation, but using only local computations. Each neuron only needs to know its own prediction error. This

is a strong candidate for how the brain might implement something functionally equivalent to backpropagation without the biologically implausible machinery.

Forward-Forward Algorithm (Hinton, 2022): Geoffrey Hinton — one of the inventors of backpropagation — proposed the Forward-Forward algorithm as a biologically plausible alternative. Instead of a forward pass followed by a backward pass, the Forward-Forward algorithm uses two forward passes: one with real data (positive pass) and one with generated 'negative' data (negative pass). Each layer learns locally to distinguish positive from negative data. This eliminates both the backward pass and the need for precisely propagated error signals.

Reward-Modulated STDP: In reinforcement learning contexts, STDP combined with a global reward signal (dopamine) can solve the credit assignment problem. STDP creates 'eligibility traces' at synapses — temporary records of which synapses were recently active together. When a reward signal arrives (via dopamine), all eligible synapses are strengthened. This allows the brain to learn from sparse, delayed rewards using only biologically plausible mechanisms.

Algorithm	Weight Transport?	Global Error?	Separate Phases?	Performance vs. Backprop
Backpropagation	Yes (problem)	Yes (problem)	Yes (problem)	Gold standard
Hebbian / STDP	No ✓	No ✓	No ✓	Limited (no credit assignment)
Feedback Alignment	No ✓	Partial	Yes	~95% of backprop
Predictive Coding	No ✓	Local errors ✓	No ✓	~Equivalent to backprop
Forward-Forward	No ✓	No ✓	No ✓	~90% of backprop (early results)
Reward-Modulated STDP	No ✓	Global reward only ✓	No ✓	Good for RL tasks



THINK ABOUT IT

The fact that predictive coding — a theory developed to explain brain function — turns out to be mathematically equivalent to backpropagation is one of the most exciting convergences in neuroscience and AI.

It suggests that the brain may be doing something functionally equivalent to backpropagation, just implemented with different biological hardware. The brain does not need weight transport or separate phases because predictive coding achieves the same result through local error signals flowing continuously between layers.

This connects directly to Module 02, where we studied the Free Energy Principle and predictive processing. The thread running through the course is becoming visible: the brain as a prediction machine, with memory as the substrate for those predictions.

? KNOWLEDGE CHECK

Test your understanding before moving on:

1. Name two reasons why standard backpropagation is considered biologically implausible.
2. What is Hebbian learning? Why is it biologically plausible but computationally limited?
3. How does feedback alignment solve the weight transport problem?
4. How does predictive coding connect to what we learned in Module 02 about the brain as a prediction machine?
5. If you were designing a neuromorphic chip that needs to learn on-device (no cloud), which biologically plausible learning algorithm would you choose, and why?

Answers are discussed in the following sections — but try to answer from memory first!

Section 8

The Big Picture — What the Brain Teaches AI About Memory

Let us step back and see the full landscape of what we have covered. The brain's memory systems offer a sophisticated, multi-faceted blueprint for AI. Some of these principles have already been adopted; others represent open frontiers.

Biological Principle	Status in AI	Example
Multiple specialized memory systems	Partially adopted	Separate parameters, context window, and external DB in RAG systems
Fast hippocampal encoding + slow cortical consolidation	Adopted (in specific domains)	Experience replay in DRL; continual learning research
Memory consolidation during sleep	Partially adopted	Offline training, generative replay
Content-addressable retrieval	Adopted	Attention mechanisms, nearest-neighbor retrieval in RAG
Pattern separation (distinct encoding)	Rarely used	Sparse representations, prototype networks
Pattern completion (retrieval from partial cues)	Adopted	Attention, Hopfield networks, associative memory
Emotional tagging of important memories	Partially adopted	Prioritized experience replay, curiosity-driven exploration
Active forgetting	Emerging	Dropout (loosely), EWC, pruning
Spatial cognitive maps	Emerging	Grid cell-like representations in navigation agents
Biologically plausible learning	Active research	Predictive coding, feedback alignment, forward-forward

 KEY TAKEAWAY: The Memory Gap

The single biggest gap between biological and artificial memory is FLEXIBILITY. Biological memory is:

- Context-sensitive: you remember different things in different situations
- Emotionally modulated: important things are remembered better
- Reconstructive: memories are rebuilt, not replayed verbatim
- Continuously updating: you learn and forget constantly
- Multi-system: different memory types for different purposes

AI memory is typically rigid, uniform, and either permanent (weights) or completely temporary (context window). Closing this gap is one of the most important challenges in AI research.

Section 9**Lab Preview — Human vs. AI Memory Performance** **LAB 06: Pattern Recognition — Human vs. AI Memory**

In this lab, you will directly compare how human memory and AI memory handle the same task.

Part A — Human Memory Experiment:

You will test your own memory using standard cognitive psychology tasks: free recall (remembering a list of words), recognition (identifying previously seen items), and serial recall (remembering items in order). You will record your performance and note which types of items you remember best (emotional words? beginning of the list? end of the list?).

Part B — AI Memory Experiment:

You will test an LSTM and a small transformer on equivalent memory tasks using Python (or Wolfram). You will measure how sequence length affects recall accuracy, test whether the models show 'primacy' and 'recency' effects (remembering the beginning and end of lists better, like humans do), and compare the error patterns of biological and artificial systems.

Part C — Analysis:

You will write a reflection comparing the strengths and weaknesses of human vs. AI memory, using the concepts from this module (working memory capacity, consolidation, pattern completion, catastrophic forgetting).

Tools: Google Colab or Wolfram Cloud. Detailed instructions will be provided in the lab document.

Section 10

Module Summary — Key Takeaways

- 1 Memory is not one thing — it is four systems** (working, episodic, semantic, procedural), each with different brain structures, different computational properties, and different AI parallels. Most AI systems only have analogues to one or two of these.
- 2 The hippocampal-cortical two-system architecture** solves the fundamental tradeoff between fast learning and stable storage. This principle — fast encoding plus slow consolidation — has been successfully adopted in AI through experience replay and continual learning strategies.
- 3 Learning has physical machinery:** LTP strengthens connections, LTD weakens them, and neuromodulators (dopamine, acetylcholine, norepinephrine) control when and how much learning occurs. The brain's learning rate is dynamic and context-dependent — unlike AI's typically fixed learning rates.
- 4 Experience replay is the clearest neuroscience-to-AI success story in this module** — hippocampal replay during sleep directly inspired the technique that made DQN, and later many deep RL systems, possible. Sleep itself is part of learning, not a pause from it.
- 5 The biological plausibility debate is not settled** — but predictive coding, feedback alignment, and the forward-forward algorithm show that the brain may implement something functionally equivalent to backpropagation using only local, biologically realistic computations.
- 6 The biggest gap between biological and AI memory is flexibility** — the brain's memory is context-sensitive, emotionally modulated, continuously updating, and reconstructive. Closing this gap is one of AI's most important open challenges.

Section 11

Bridge to Module 07 — Attention and Consciousness in AI



WHAT COMES NEXT

Module 06 explored how the brain stores and retrieves information — the memory systems that make learning possible.

Module 07 takes us deeper: once the brain has sensory information (Module 05) and can store it (Module 06), how does it **DECIDE** what to focus on? How does it select what reaches conscious awareness? And does AI need consciousness to be truly intelligent?

Module 07 will explore:

- Biological attention: spatial, feature-based, and executive attention circuits
- The relationship between attention and consciousness (Global Workspace Theory, Integrated Information Theory)
- Attention mechanisms in AI: from soft attention to multi-head self-attention
- The hard problem of consciousness — and whether it matters for AI
- Ethical implications: if AI models have attention, do they have experience?

Teaser: There is a theory that consciousness is what happens when information is globally broadcast across the brain — made available to all brain systems simultaneously. Transformers do something strikingly similar with their attention mechanism. Does that make them conscious? Almost certainly not — but understanding why not will teach you something profound about both brains and machines.

Section 12

Glossary

Acetylcholine: A neuromodulator released during attention and alertness that enhances LTP and new learning. Destroyed in Alzheimer's disease.

Catastrophic Forgetting: The tendency of neural networks to completely lose previously learned information when trained on new data. Also called catastrophic interference.

Cell State: The internal memory highway in an LSTM cell that carries information through time with minimal modification.

Complementary Learning Systems (CLS): The theory that the brain uses two complementary learning systems — the fast-learning hippocampus and the slow-learning cortex — to balance rapid acquisition with stable long-term storage.

Content-Addressable Memory: A memory system where stored items are retrieved based on their content (similarity to a query) rather than their storage location. The hippocampus and transformer attention both implement this.

Dentate Gyrus: Part of the hippocampal circuit responsible for pattern separation — creating distinct memory representations for similar experiences.

Differentiable Neural Computer (DNC): A memory-augmented neural network from DeepMind that can learn to read, write, and navigate data structures in external memory.

Dopamine: A neuromodulator that signals reward prediction error — surprise about outcomes. Acts as a learning gate: when released, it enhances LTP at recently active synapses.

Elastic Weight Consolidation (EWC): An algorithm that prevents catastrophic forgetting by protecting weights that are important for previously learned tasks.

Episodic Memory: Memory for personally experienced events including their context (what, where, when). Dependent on the hippocampus. Learned in one shot.

Experience Replay: An AI technique inspired by hippocampal replay where past experiences are stored in a buffer and randomly sampled during training to stabilize learning.

Feedback Alignment: A biologically plausible alternative to backpropagation that uses random, fixed feedback weights instead of transposed forward weights.

Forward-Forward Algorithm: A learning algorithm proposed by Hinton (2022) that replaces the backward pass with two forward passes (positive and negative data).

Grid Cells: Neurons in the entorhinal cortex that fire in hexagonal patterns as an animal moves through space, providing a coordinate system for navigation.

Hebbian Learning: The principle that 'neurons that fire together wire together' — synapses are strengthened when pre- and postsynaptic neurons are active simultaneously.

Long-Term Depression (LTD): The weakening of a synaptic connection following low-frequency stimulation. The biological 'eraser' that complements LTP.

Long-Term Potentiation (LTP): The strengthening of a synaptic connection following high-frequency stimulation. The primary cellular mechanism of learning, discovered by Bliss and Lømo (1973).

LSTM (Long Short-Term Memory): A recurrent neural network architecture with gating mechanisms that allow information to persist over long sequences.

Memory Consolidation: The process of transferring memories from temporary hippocampal storage to permanent cortical storage, primarily occurring during sleep.

Neural Turing Machine (NTM): A neural network with external memory that can learn to perform algorithmic tasks by reading from and writing to a memory matrix.

Neuromodulator: A chemical signal (like dopamine, acetylcholine, or norepinephrine) that does not directly cause neurons to fire but changes how responsive they are to other inputs — controlling when and how much learning occurs.

NMDA Receptor: A type of receptor on neurons that acts as a coincidence detector — it only opens when the presynaptic AND postsynaptic neurons are both active, making it the molecular basis of Hebbian learning.

Norepinephrine: A neuromodulator released during arousal and novelty that enhances alertness and learning. Plays a role in the explore-versus-exploit tradeoff.

Pattern Completion: Retrieving a complete memory from a partial cue, performed by the CA3 region of the hippocampus.

Pattern Separation: Creating distinct neural representations for similar experiences, performed by the dentate gyrus.

Place Cells: Hippocampal neurons that fire when an animal is in a specific location, forming an internal map of the environment.

Predictive Coding: A framework where each brain layer predicts the activity of the layer below, and only prediction errors are propagated. Can implement learning equivalent to backpropagation.

Procedural Memory: Memory for skills and habits (how to ride a bike, type on a keyboard). Supported by the basal ganglia and cerebellum. Requires thousands of repetitions.

RAG (Retrieval-Augmented Generation): An AI architecture that combines a language model with an external database, retrieving relevant information before generating responses.

REM Sleep: Rapid Eye Movement sleep — the sleep stage associated with dreaming, procedural memory consolidation, emotional processing, and creative recombination of memories.

Semantic Memory: Memory for general knowledge and facts, independent of the personal context in which they were learned.

Slow-Wave Sleep: Deep Non-REM sleep characterized by large, slow brain oscillations. The stage when hippocampal replay occurs, consolidating episodic and semantic memories.

Trisynaptic Circuit: The main internal pathway of the hippocampus: entorhinal cortex → dentate gyrus → CA3 → CA1.

Working Memory: The ability to hold and manipulate a small amount of information temporarily. Supported by the prefrontal cortex.

Section 13

References and Further Reading

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Online Resources

- Neuroscience Online (UTHealth) — Chapter on Memory Systems: nba.uth.tmc.edu/neuroscience
- 3Blue1Brown — 'But what is a neural network?' (visual explanation of backpropagation): youtube.com/3blue1brown
- DeepMind Blog — DQN and Experience Replay: deepmind.com/research
- Wolfram Demonstrations — Hopfield Network Memory: demonstrations.wolfram.com
- Yannic Kilcher — Video explanation of the Forward-Forward Algorithm: youtube.com/yannickilcher